

BIVAR CAPITAL

VC2 Value Composite

Strategy Methodology & Proof of Concept

A systematic value investing strategy that ranks stocks using six fundamental metrics, backed by decades of academic research.

Data Source: Sharadar / Nasdaq Data Link

Universe: US Equities | Market Cap > \$200M | Excluding financials & negative earnings

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1. What is the VC2 Value Composite?

The VC2 (Value Composite 2) is a multi-factor value strategy originally developed by James O'Shaughnessy in *What Works on Wall Street*. It ranks every stock in the market using six fundamental value metrics, combines them into a single composite score, and selects the cheapest stocks.

The key insight: no single valuation metric works all the time. P/E misses capital-light businesses. P/B misses intangibles. EV/EBITDA misses leverage. By combining six metrics, the composite captures value across different business models and market conditions.

O'Shaughnessy's backtests showed that the VC2 strategy delivered approximately 18-21% CAGR over 40+ years of US market data, significantly outperforming both the S&P 500 (~10.9% CAGR) and any single-metric value strategy.

2. The Six Metrics

Each stock is ranked on six fundamental metrics. Lower is cheaper (better) for the first five. Higher is better for Shareholder Yield.

METRIC 1: PRICE-TO-EARNINGS (P/E)

Share price divided by earnings per share. The most widely used valuation metric. Lower P/E = cheaper relative to earnings. Limitation: can be distorted by one-time charges or accounting choices.

METRIC 2: PRICE-TO-SALES (P/S)

Share price divided by revenue per share. Revenue is harder to manipulate than earnings, making P/S more stable. Lower P/S = cheaper relative to sales. Works well for companies with volatile earnings.

METRIC 3: PRICE-TO-BOOK (P/B)

Share price divided by book value per share. Measures how much you're paying for the company's net assets. Lower P/B = cheaper. Limitation: less meaningful for asset-light or intangible-heavy businesses. We exclude financial stocks where P/B has different implications.

METRIC 4: PRICE-TO-FREE-CASH-FLOW (P/FCF)

Share price divided by free cash flow per share. FCF is the cash a company generates after capital expenditures — the real cash available to shareholders. Lower P/FCF = more cash generation per dollar invested. Often considered the purest measure of value.

METRIC 5: EV/EBITDA

Enterprise Value (market cap + debt - cash) divided by EBITDA (earnings before interest, taxes, depreciation & amortization). This metric accounts for capital structure — a company with heavy debt will have a higher EV. Lower EV/EBITDA = cheaper when accounting for debt.

METRIC 6: SHAREHOLDER YIELD

Dividend Yield + Buyback Yield. This is the total cash returned to shareholders, not just dividends. Many companies return more cash through buybacks than dividends. A company with 1% dividend

yield but 5% buyback yield has a 6% Shareholder Yield — far more attractive than the dividend alone suggests.

Dividend Yield = Annual dividends per share / Share price

Buyback Yield = Net share repurchases / Market cap (calculated from net cash from common equity transactions)

Higher Shareholder Yield = better (more cash returned to you).

3. How the Composite Score Works

The VC2 composite ranking process:

Step 1 — Universe. Start with all US stocks above \$200M market cap.

Step 2 — Exclude. Remove financial sector stocks (banks, insurers — P/B and EV/EBITDA aren't meaningful). Remove stocks with negative earnings.

Step 3 — Rank. For each of the six metrics, rank every stock from cheapest (percentile 0) to most expensive (percentile 100).

Step 4 — Composite. Average the six percentile ranks into a single VC2 score. Lower score = cheaper across all dimensions.

Step 5 — Select. Buy the 25 stocks with the lowest composite score. Equal weight (4% each).

Step 6 — Rebalance. Repeat annually. Sell all positions, re-run the ranking, buy the new top 25.

Why percentile ranks instead of raw values?

Raw P/E of 10 vs EV/EBITDA of 8 are not comparable. By converting each metric to a percentile rank (where does this stock sit relative to all other stocks?), we can meaningfully average across different scales. A stock in the 5th percentile for P/E and 10th percentile for P/S is consistently cheap, regardless of the absolute numbers.

4. Proof of Concept — Live Results

Using Sharadar (Nasdaq Data Link) data with the free tier (limited to ~30 large-cap tickers), we ran the full VC2 ranking to validate the methodology. Below are the results.

#	Ticker	VC2	P/E	P/S	P/B	P/FCF	EV/EB	SH Yield
1	XOM	10.9	11.0	1.15	1.94	11.8	5.5	8.2%
2	VZ	12.0	13.6	1.18	1.72	8.5	8.2	7.0%
3	CVX	13.4	13.2	1.40	1.75	13.8	6.3	9.2%
4	TRV	19.0	14.5	1.05	1.75	5.7	13.6	4.1%
5	CAT	30.7	14.6	2.25	7.72	14.3	11.5	5.0%
6	AXP	31.6	16.3	2.46	4.87	8.1	11.6	3.9%
7	CSCO	32.3	16.8	3.72	4.79	11.2	12.2	5.0%
8	JPM	32.6	10.3	3.11	1.50	38.5	9.8	4.4%
9	JNJ	38.0	10.7	4.43	5.49	21.8	8.6	4.0%
10	IBM	40.1	19.9	2.41	6.63	12.0	13.7	3.8%

Interpretation: XOM, VZ, CVX rank as the cheapest — low P/E, low P/S, low EV/EBITDA, and high Shareholder Yield (8-9%). These are classic deep value stocks returning significant cash to shareholders. The ranking correctly identifies value across multiple dimensions.

Shareholder Yield Breakdown (Top 5):

Ticker	Div Yield	Buyback Yield	Shareholder Yield
CVX	4.0%	5.2%	9.2%
XOM	3.7%	4.5%	8.2%
DD	1.9%	6.0%	7.9%
GS	2.7%	4.6%	7.3%
VZ	7.0%	0.0%	7.0%

Note: CVX returns 9.2% of its market cap to shareholders annually (4% dividends + 5.2% buybacks). Using only dividend yield would miss more than half of the cash return.

5. Data Source & Quality

Sharadar Core US Equities Bundle (Nasdaq Data Link)

Feature	Detail
Coverage	21,000+ US companies
History	Since January 1998 (28 years)
Delisted	Yes — survivorship-bias free
Fundamentals	112 fields per company per period
Frequency	Daily prices, Quarterly/Annual fundamentals
Key Fields	P/E, P/S, P/B, EV/EBITDA, FCF, dividends, buybacks, market cap
Update	Daily

Why Survivorship-Bias Free Data Matters

Most free data sources (Yahoo Finance, Google Finance) only contain stocks that exist today. Companies that went bankrupt, were acquired, or delisted are missing. This inflates backtest results because you're only testing against survivors — the stocks that didn't fail.

Sharadar includes all 21,000+ companies that were ever listed on US exchanges, including those that were subsequently delisted. This means our backtests reflect reality: some stocks in our portfolio will fail, and the results account for that.

6. Next Steps

Full Backtest (1998-2026) — Run the VC2 strategy on the full 21,000 stock universe with annual and monthly rebalancing. Calculate CAGR, Sharpe ratio, max drawdown, win rate.

Momentum Strategy — Implement intermediate momentum (Novy-Marx 2012, months 7-12) as a standalone strategy.

Momentum + Value (Trending Value) — Combine VC2 with momentum: take the cheapest 10% by VC2 score, then rank by 6-month momentum. Buy the top 25.

Strategy Comparison — Compare all three strategies vs S&P; 500, including during crisis periods (2008, 2020, 2022).

Live Deployment — Daily ranking updates on bivarcapital.com using the same methodology.

Bivar Capital — bivarcapital.com

Not financial advice. Past performance does not guarantee future results.

Data: Sharadar / Nasdaq Data Link. Methodology: O'Shaughnessy, What Works on Wall Street.